

Database Design Document: Wayfinder

1. Introduction

This document outlines the database schema for the Wayfinder bus tracking and digital verification system. The database is designed to support a three-tier Progressive Web App (PWA) architecture featuring real-time WebSocket-based tracking and secure digital pass verification. By eliminating the need for dedicated hardware, the system relies on this central data architecture to manage high-fidelity navigation and credentialing tools.

2. Entity-Relationship Overview

The system manages three primary user roles: Students, Drivers, and Admins. It facilitates operational logistics including route management, real-time vehicle location updates streamed via WebSockets, and fraud-proof digital pass verification. The architecture ensures referential integrity between transit operations and user credentials.

3. Data Schema

3.1 Users Table

Stores core identity information for all stakeholders. Sessions are authenticated using JSON Web Tokens (JWT) based on these records.

Column Name	Data Type	Constraints
user_id	UUID	Primary Key
name	String	Not Null
email	String	Unique, Not Null
password_hash	String	Secure Hash

Column Name	Data Type	Constraints
role	Enum	'admin', 'student', 'driver'
created_at	Timestamp	Default: Current

3.2 Buses Table

Defines the physical fleet managed by the Administrative Dashboard.

Column Name	Data Type	Constraints
bus_id	UUID	Primary Key
license_plate	String	Unique, Not Null
assigned_driver_id	UUID	Foreign Key (Users), Nullable
current_status	Enum	'active', 'maintenance', 'idle'

3.3 Routes Table

Manages the logical transit paths within the system.

Column Name	Data Type	Constraints
route_id	UUID	Primary Key
route_name	String	Not Null
description	Text	Nullable
is_active	Boolean	Default: True

3.4 Route_Stops Table

A join table facilitating many-to-many relationships between routes and geographic stops.

Column Name	Data Type	Constraints
id	PK (Serial)	Primary Key
route_id	UUID	Foreign Key (Routes)
stop_name	String	Not Null
latitude	Decimal	Precision 9,6
longitude	Decimal	Precision 9,6

3.5 Student_Passes Table

Supports the digital pass module where secure, encrypted QR codes are generated for boarding.

Column Name	Data Type	Constraints
pass_id	UUID	Primary Key
student_id	UUID	Foreign Key (Users)
status	Enum	'active', 'expired', 'pending'
expiry_date	Date	Not Null
qr_code_hash	String	Hashed for security

3.6 Trips Table

Tracks operational history and links drivers to specific routes for reporting and heat map generation.

Column Name	Data Type	Constraints
trip_id	UUID	Primary Key
bus_id	UUID	Foreign Key (Buses)
driver_id	UUID	Foreign Key (Users)
start_time	Timestamp	Not Null
route_id	UUID	Foreign Key (Routes)

3.7 Live_Location Table

Facilitates sub-second coordinate streaming via WebSockets for real-time map visualization.

Column Name	Data Type	Constraints
id	PK (Serial)	Primary Key
bus_id	UUID	Foreign Key (Buses)
latitude	Decimal	Precision 9,6
longitude	Decimal	Precision 9,6
updated_at	Timestamp	Default: Current

4. Security & Data Integrity

- **PII Encryption:** Sensitive student data and location history must be encrypted at rest and in transit to comply with the DPDP Act 2023.
- **Verification:** The `qr_code_hash` utilizes hashed user IDs and timestamps to prevent forgery or duplication during the real-time scan-and-validate process.
- **Constraints:** Strict foreign key constraints and Role-Based Access Control (RBAC) are applied at the database level to ensure data security and maintain referential integrity.

- **Performance:** The schema is optimized for sub-second latency, targeting database response times under 2 seconds for high-volume concurrent location packets.